Trigger	Answer
Dot product meaning	$u \cdot v = \|u\|\|v\| \cos\theta$
Matrix product valid	inner dimensions match
Matrix product shape	$(p \times q)(q \times s) \to p \times s$
Remove hidden layer from MLP	linear mapping
Variance term	predictions change across training sets
Low train, high val error	overfitting
Sparse weights	L_1 / lasso
CNN fewer params	weight sharing across positions
k-means update	mean of assigned points
k-means fails on	stretched / non-convex / unequal-density clusters
1st principal component	direction of max variance
Curved boundary, few assumptions	k-NN
Global assumptions	overall shape restriction on whole dataset
Parallelize	train independent pieces simultaneously
Ensemble averaging helps most	models make different errors

Trigger	Answer
UCB choosing weakly tried arm	exploration bonus / uncertainty
Greedy vs ϵ-greedy vs UCB	best mean / random sometimes / best mean + bonus
Well-calibrated probs	predicted p correct about p% of time
SVM key fact	remove non-support vectors: no change
95/5 class split + 95% acc	misleading, use recall / precision / F1
Negative R^2	worse than mean predictor
Same acc, different CE loss	different confidence quality
t-SNE reliable info	local neighbors only
Spurious correlation	predictive maybe, causal no
Entropy / IG / Gini	impurity / entropy drop / impurity alternative
Connectivity-based grouping	linked neighbors / dense regions, not centroids
Curse of dimensionality	distances less meaningful, hurts k-NN
Discounted rewards	earlier reward counts more
Ring-shaped classes good	k-NN, RBF SVM, trees, nets
Ring-shaped classes bad	logistic, linear SVM, OLS

Linear Algebra / Setup

- Vectors $x \in \mathbb{R}^d$, matrix $X \in \mathbb{R}^{n \times d}$.
- Dot product: $u \cdot v = \sum_i u_i v_i = \|u\|\|v\| \cos\theta$.
- Norms: $\|x\|_2 = \sqrt{\sum_i x_i^2}$, $\|w\|_1 = \sum_j |w_j|$.
- Matrix-vector: $\hat{y} = Xw$, and $\hat{y}_i = x^{(i)} \cdot w$.
- If $A \in \mathbb{R}^{p \times q}, B \in \mathbb{R}^{r \times s}$, then AB exists iff $q = r$.
- If valid, AB has shape $p \times s$.
- Transpose rule: $(AB)^T = B^T A^T$.

Regression / GD

- Linear model: $\hat{y} = w^T x + b$.
- MSE: $\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$.
- OLS: $\hat{w} = (X^T X)^{-1} X^T y$.
- GD: $w^{(t+1)} = w^{(t)} - \eta \nabla_w L(w^{(t)})$.
- η too large: overshoot, oscillation, divergence.
- Negative R^2: worse than predicting the mean.

Bias / Variance / Fit

- High bias: too simple, underfits.
- High variance: predictions change a lot across training sets.
- Underfitting: high train and high validation error.
- Overfitting: low train error, much higher validation/test error.
- Train = fit parameters, validation = tune hyperparameters, test = final estimate.

Regularization

- Generic form: loss $+\lambda$(penalty).
- Ridge / L_2: RSS $+ \lambda \sum_j w_j^2$.
- Lasso / L_1: RSS $+ \lambda \sum_j |w_j|$.
- L_1 gives sparsity and feature selection.
- L_2 shrinks coefficients but usually not exactly to zero.
- Larger λ: more bias, less variance, simpler model.

Binary / Multiclass Models

- Linear regression output: $\hat{y}_i = w^T x_i + b$.
- Sigmoid: $\sigma(z) = \frac{1}{1+e^{-z}}$.
- Binary class prob: $p_i = \sigma(w^T x_i + b)$.
- Logit: $\log \frac{p}{1-p} = w^T x + b$.
- Multiclass score: $s_{ik} = w_k^T x_i + b_k$.
- Softmax: $p_{ik} = \frac{e^{s_{ik}}}{\sum_{j=1}^K e^{s_{ij}}}$.
- Predict class: $\hat{y}_i = \arg\max_k s_{ik}$.

Loss / Accuracy / Calibration

- Squared loss: $\frac{1}{n} \sum_i (y_i - \hat{y}_i)^2$.
- Binary CE: $-\frac{1}{n} \sum_i [y_i \log p_i + (1 - y_i) \log(1 - p_i)]$.
- Multiclass CE: $-\frac{1}{n} \sum_i \sum_k y_{ik} \log p_{ik}$.
- Accuracy: $\frac{TP+TN}{TP+TN+FP+FN}$.
- Precision: $\frac{TP}{TP+FP}$.
- Recall: $\frac{TP}{TP+FN}$.
- F1: $\frac{2PR}{P+R}$.
- Same accuracy, different loss: same right/wrong, different confidence.
- Well-calibrated: if model predicts p, it is right about p% of the time.

Imbalanced Data

- 95% class A, 5% class B: always predicting A gives 95% accuracy.
- Accuracy can hide complete failure on minority class.
- Better metrics: confusion matrix, precision, recall, F1, class-specific recall, PR curve.
- Phrase: "model may just predict majority class every time."

k-NN

- Predict from nearby points, no explicit training.
- Good for highly curved / irregular boundaries.
- Minimal global assumptions.
- Sensitive to scaling and high dimension.
- Curse of dimensionality: in high dimension, distances become less informative.

Neural Nets / CNNs

- Hidden-layer MLP: nonlinear map. Remove hidden layer $\Rightarrow$ linear mapping.
- ReLU: $\text{ReLU}(x) = \max(0, x)$.
- Sigmoid derivative: $\sigma'(x) = \sigma(x)(1 - \sigma(x))$.
- Tanh derivative: $(\tanh x)' = 1 - \tanh^2 x$.
- CNNs use fewer parameters because filters are reused across spatial locations.
- Key phrase: weight sharing.

Decision Trees

- Recursive partitioning model, leaf predicts majority class.
- Criteria: entropy, information gain, Gini.
- Entropy: $H = -\sum_k p_k \log p_k$.
- Gini: $1 - \sum_k p_k^2$.
- Information gain = parent entropy $-$ weighted child entropy.
- Entropy and Gini often choose similar splits.
- Entropy can be a bit more sensitive near pure nodes.
- Bigger tree: lower bias, higher variance.

Tree Misclassification / Pruning

- Leaf mistakes = total in leaf $-$ majority-class count.
- Tree misclassification rate = total mistakes / total samples.
- Count leaf nodes only, not internal splits.
- Cost-complexity pruning: $\text{Cost}(T) = \text{error}(T) + \lambda L(T)$.
- Compare old cost vs new cost directly.
- Prune if new cost $<$ old cost.
- Quick rule: prune if error increase $< \lambda \cdot$ leaves removed.

Bagging / RF / Boosting

- Bagging: bootstrap many trees, average or vote. Main effect: lower variance.
- Random forest = bagging + random subset of features at each split.
- RF usually lower variance than plain bagging because trees are less correlated.
- RF parallelizes well: trees are independent.
- Boosting: sequentially trains weak learners to correct earlier mistakes.
- Boosting parallelizes poorly: later trees depend on earlier trees.
- Boosting is more likely to overfit as # trees increases.
- Averaging helps most when models make different errors.

SVM

- Finds large-margin decision boundary.
- Support vectors are the points that determine the boundary.
- Non-support vectors do not affect final boundary.
- Remove only non-support vectors $\Rightarrow$ classifier unchanged.
- Linear SVM: linear boundary.
- RBF SVM: nonlinear, good for curved boundaries.

Unsupervised Learning

- No labels, discover structure.
- Applications: customer segmentation, anomaly detection, grouping similar users / products.
- Phrase: "there are no ground-truth labels, so the goal is to discover natural structure."

k-Means

- Repeat: assign each point to nearest centroid, then update centroid to cluster mean.
- Update formula: $\mu_k = \frac{1}{|C_k|} \sum_{x_i \in C_k} x_i$.
- Works best for compact, spherical, similarly sized clusters.
- Struggles on stretched, curved, non-convex, unequal-density groups.
- Centroid-based, not connectivity-based.

Connectivity-Based Clustering

- Group points because they are linked through neighbors / dense regions.
- Think linked structure, not nearest centroid.
- Good for irregular shapes.
- Hierarchical agglomerative is naturally more connectivity-based than k-means.

PCA

- Principal components are orthogonal directions of high variance.
- First PC = direction of maximum projected variance.
- Larger eigenvalue λ means more variance explained.
- PCA is for variance compression, not class separation.
- Safe interpretation: top PCs explain more variance.

t-SNE

- Nonlinear visualization method for high-dimensional data.
- Preserves local neighborhood structure.
- Does not reliably preserve global distances or cluster spacing.
- Safe claim: close in 2D $\Rightarrow$ likely neighbors in original space.
- Unsafe claim: far apart in 2D $\Rightarrow$ must be far apart originally.

Spurious Correlation / Causality

- Predictive relationship may be non-causal.
- Examples: wolf vs husky with snow, drownings vs ice cream sales.
- Problem: model learns shortcut and generalizes poorly when environment changes.
- Phrase: "predictive in observed data, but not causal."

Bandits / RL

- Greedy: choose highest current estimated reward.
- ϵ-greedy: greedy most of the time, random with probability ϵ.
- UCB: choose high mean reward + uncertainty bonus.
- Weakly tried arm chosen despite worse mean $\Rightarrow$ think UCB.

Discounted Reward

- Return: $G = r_0 + \gamma r_1 + \gamma^2 r_2 + \cdots$.
- With $\gamma = 0.5$: one step later gets 0.5, two steps later 0.25, three steps later 0.125.
- Tiny example: 1 at $t+1$ and 1 at $t+2$ gives $0.5 + 0.25 = 0.75$.

Routing Triggers

- Numeric target $\to$ linear regression
- Binary label $\to$ logistic regression
- Multiclass label $\to$ softmax classifier
- No labels, find compact groups $\to$ k-means
- No labels, preserve variance $\to$ PCA
- No labels, inspect local neighborhoods in 2D $\to$ t-SNE
- Highly curved boundary $\to$ k-NN, RBF SVM, trees, nets
- Strong tabular classifier, independent trees $\to$ random forest
- Strong tabular classifier, sequential correction $\to$ boosting
- Need interpretability $\to$ decision tree
- Image spatial patterns $\to$ CNN
- Explore vs exploit $\to$ bandits / UCB
- Honest probabilities $\to$ calibration

Graph / Metric Diagnostics

- Train loss low, val loss high $\to$ overfitting
- Train and val both high $\to$ underfitting
- Loss oscillates wildly $\to$ learning rate too large
- $R^2 = 1$ perfect fit, $R^2 = 0$ mean baseline, $R^2 < 0$ worse than baseline
- High acc on imbalanced data $\to$ check recall / precision / F1
- Same acc, lower CE $\to$ better probability outputs

Common Confusion Traps

- PCA $\neq$ clustering
- t-SNE $\neq$ reliable global distances
- Bagging $\neq$ boosting
- Calibration $\neq$ accuracy
- $L_1 \neq L_2$
- Logistic regression $\neq$ linear regression
- Random forest $\neq$ single decision tree
- Support vectors $\neq$ all training samples
- Connectivity-based clustering $\neq$ centroid-based clustering

Ultra-Short Memory Hooks

- Dot product = angle
- Matrix product = inner match, outer shape
- Variance = instability
- Overfit = train good, val bad
- L_1 = sparse
- CNN = shared weights
- k-means = means
- PCA = max variance
- RF = bagging + random features
- Boosting = sequential correction
- SVM = support vectors matter
- UCB = uncertainty bonus
- Calibration = honest probabilities
- t-SNE = local only
- High dimension = distances weaken

Model Summary Table

Model	Type	Loss / Rule	Best for	Notes / traps
Linear regression	Sup	MSE	numeric prediction	linear output $\hat{y} = w^T x + b$
Logistic regression	Sup	binary CE	binary class, linear boundary	sigmoid output, poor for ring data
Softmax classifier	Sup	multiclass CE	multiclass labels	one score per class
k-NN	Sup	distance + vote	small data, curved boundary, few assumptions	suffers in high dimension, scaling matters
Decision tree	Sup	Gini / entropy / IG	interpretable rule splits	high variance, can overfit
Random forest	Sup	ensemble vote	strong tabular prediction	independent trees, parallelizable, lower variance
Boosted trees	Sup	sequential correction	very strong tabular prediction	less parallelizable, can overfit sooner
Linear SVM	Sup	max-margin	linear separation	support vectors matter
RBF SVM	Sup	max-margin in kernel space	nonlinear curved boundary	good for ring-shaped classes
Neural net / MLP	Sup	GD on chosen loss	flexible nonlinear mapping	unstable if learning rate too large
CNN	Sup	GD on chosen loss	image / spatial structure	fewer params due to weight sharing
k-means	Unsup	minimize within-cluster SSE	compact spherical clusters	centroid = mean, poor for crescents
Hierarchical clustering	Unsup	linkage rule	connectivity / nested grouping	not centroid-based
PCA	Unsup	maximize projected variance	dimension reduction	linear, variance not labels
t-SNE	Unsup	neighborhood-preserving map	2D visualization	local structure only, global spacing unreliable
Bandits / UCB	RL	reward + exploration	explore/exploit decisions	UCB favors uncertain arms

Metrics / Routing Table

Situation / goal	Route to	Why
Need numeric prediction	linear regression	continuous target
Need binary class probability	logistic regression	sigmoid gives binary probability
Need multiclass probabilities	softmax classifier	one probability per class
No labels, preserve variance	PCA	linear dimension reduction by variance
No labels, group similar customers	k-means or hierarchical clustering	segmentation without labels
Need 2D visualization of local neighborhoods	t-SNE	local neighborhoods preserved better than global spacing
Need strong tabular classifier with independent trees	random forest	trees train independently, easy to parallelize
Need very strong tabular accuracy via sequential correction	boosting	each learner fixes earlier mistakes
Boundary highly curved, dimension modest	k-NN or RBF SVM	flexible nonlinear decision boundary
Need interpretability	decision tree	explicit if-then split structure
Need honest probabilities	calibration / cross-entropy awareness	probability quality, not just accuracy
Severely imbalanced data	precision, recall, F1, confusion matrix	accuracy may be misleading
Feature predicts but is non-causal	spurious correlation warning	may fail when environment changes
Poorly tried bandit arm chosen	UCB	uncertainty bonus encourages exploration
High-dimensional distance-based setting	caution with k-NN	curse of dimensionality weakens distance meaning

Tree / Bandit Mini Table

Concept	Key fact
Entropy	$H = -\sum_k p_k \log p_k$
Gini	$1 - \sum_k p_k^2$
Information gain	parent entropy − weighted child entropy
Leaf misclassification	leaf total − majority class count
Tree cost-complexity	$\text{Cost}(T) = \text{error}(T) + \lambda L(T)$
Pruning check	prune if new cost $<$ old cost
Greedy bandit	choose current best mean reward
ϵ-greedy	greedy usually, random sometimes
UCB	choose estimated reward + uncertainty bonus
Discounted return	$G = r_0 + \gamma r_1 + \gamma^2 r_2 + \cdots$